

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First Semester of B. Tech. (CL/ME/EE) Theory Examination****November-December 2018****ME141 Engineering Graphics****Date: 26/11/2018, Monday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and Section II must be attempted in separate drawing sheets.
3. Make suitable assumptions and draw neat figures wherever is required.
4. Figures drawn in the question paper are not to the scale.
5. Show all the necessary dimensions in the solution.
6. Figures to right indicate the full marks.
7. Stencil work is not compulsory.

SECTION – I**Q - 1 Do as directed.**

1. Drafter is used to draw _____ lines on the sheet. **[01]**
 a) Horizontal b) Vertical c) Inclined d) All of these
2. Engineering graphics is the universal language and tool of communication for engineers. **[01]**
 True or False?
3. The size of title block as recommended by B.I.S. is _____. **[01]**
 a) 185 mm × 65 mm b) 135 mm × 65 mm c) 185 mm × 50 mm d) 135 mm × 50 mm
4. In engineering scale, $R.F. = \frac{?}{\text{Actual length of object}}$ **[01]**
 a) Length of the span b) Length of the drawing c) Both a & b d) None of these
5. Orthographic projection is used to draw 2D views of 3D object by _____. **[01]**
 a) Mechanical Engineers b) Civil Engineers c) Electrical Engineers d) All of these
6. In first angle projection system, right hand side view is located at the _____ side of the front view. **[01]**
 a) Top b) Bottom c) Left d) Right
7. The front view of a point is observed below XY line, when the point is in _____. **[01]**
 a) 3rd quadrant b) 4th quadrant c) Both a & b d) None of these

Q – 2.a Two points X & Y are 100 mm apart. A point Z is 75 mm from X and 45 mm from Y. **[06]**
 Draw an ellipse passing through points X, Y, and Z. Draw normal and tangent at any point on the curve.

Q – 2.b A room 6 m × 5 m × 4 m high has a light bracket above the center of the long wall and 1 m below the ceiling. The light bulb is 0.3 m away from the wall. The switch for the light **[06]**

is on an adjacent wall, 1.5 m above the floor and 1m from the other longer wall. Determine graphically the shortest distance between the bulb and the switch.

OR

- Q – 2.b** The length of the top view of a line parallel to the VP and inclined at 45° to the HP is 50 mm. One end of the line is 12 mm above the HP and 25 mm in front of VP. Draw the projection of a line and determine its true length. [06]
- Q – 2.c** A point E is 25 mm above HP and 50 mm away from VP. Draw the projections of a point E. Also, draw the projection of point E when it rests in HP and VP. [03]
- Q – 3.a** **Figure 1** shows pictorial view of an object. Draw the following views: (1) Front View, and (2) Top View. Dimensions shown in the figure are in mm. (**Figure 1** is on Page 3) [08]
- Q – 3.b** A cylinder having diameter of base 50 mm and height of axis 70 mm is resting on HP on its base. Draw the front view, top view of cylinder. Also, draw the development of cylinder. [05]

OR

- Q – 3.b** A square pyramid having side of base 30 mm and height of axis 60 mm is resting on HP in such a way that all the edges of base become equally inclined to VP. Draw the front view, top view of square pyramid. Also, draw the development of a square pyramid. [05]

SECTION – II

Q - 4 Do as directed.

- Classify the solids. [02]
- A plate is perpendicular to VP and inclined at 30° to HP, then the angle made by a plate with PP is _____. [01]
a) 30° b) 45° c) 60° d) 90°
- If the front view and top view of a square plate appear as a line. Then, the square plate becomes _____ to VP. [01]
a) Parallel b) Perpendicular c) Both a & b d) None of these
- Enlist the applications of plane. [01]
- In isometric scale, true length of an object is drawn at _____. [01]
a) 30° b) 45° c) 60° d) 90°
- The isometric view of a circle is drawn by concentric circle method. True or False? [01]

- Q – 5.a** A hexagonal prism having the side of hexagon 30 mm and height of 70 mm is resting on HP on one of the edge of its hexagonal base in such a way that, the edge of base is at 60° to the VP and the base is at 30° to the HP. [09]

OR

- Q – 5.a** A square prism of 40 mm base side and axis of 80 mm has its base on the HP and its faces equally inclined to VP. It is cut by a plane, perpendicular to VP, inclined 60° to HP and passing through a point on the axis at 55 mm above base. Draw projection of solid. [09]

Draw FV, section TV and True shape of section.

- Q – 5.b** A regular hexagonal plate 40 mm side is resting on one of its corners in HP. Its surface is inclined at 45° to the HP and the top view of the diagonal through the corner, which is in the HP makes an angle of 60° with the VP. Draw the projections of hexagonal plate. [06]

- Q – 6.a** **Figure 2** shows the orthographic projections of an object in first angle projection system. [08]
Draw its isometric view. Dimensions shown in the figure are in mm. (**Figure 2** is on Page 3)

OR

- Q – 6.a** Draw the isometric view of a pentagonal prism. Consider the edge of base as 50 mm and height of the axis as 75 mm for the isometric drawing of a pentagonal prism. [08]

- Q – 6.b** Discuss the benefits of CAD software. [05]

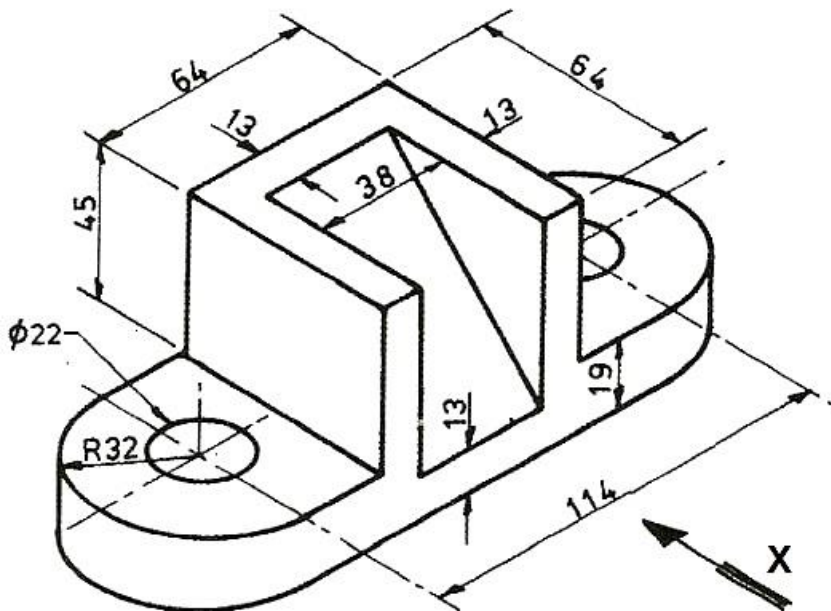
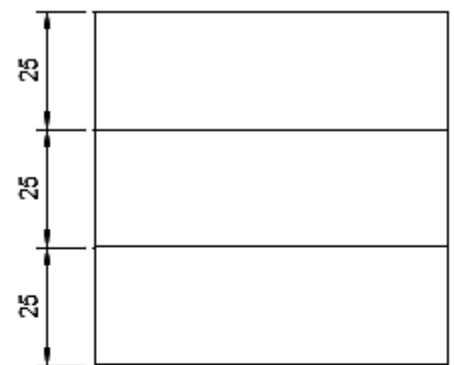
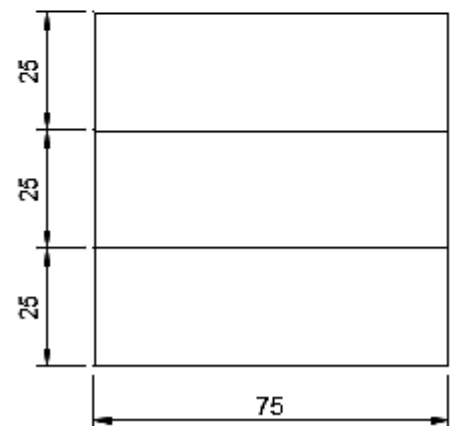


Figure 1



FV



TV

Figure 2
